

Brekfuz

The learning layer of
the internet.

We've seen

Textbooks become PDFs.

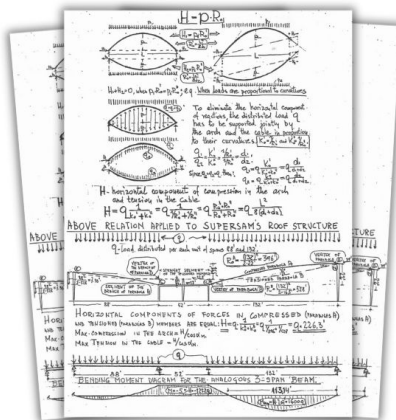
Lectures become YouTube videos.

Classrooms become screens.

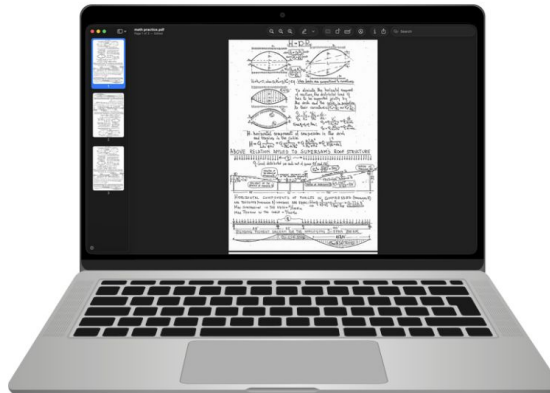
Students become spectators.

“EdTech” promised revolution, yet it only delivered **replication**.

Pre-EdTech



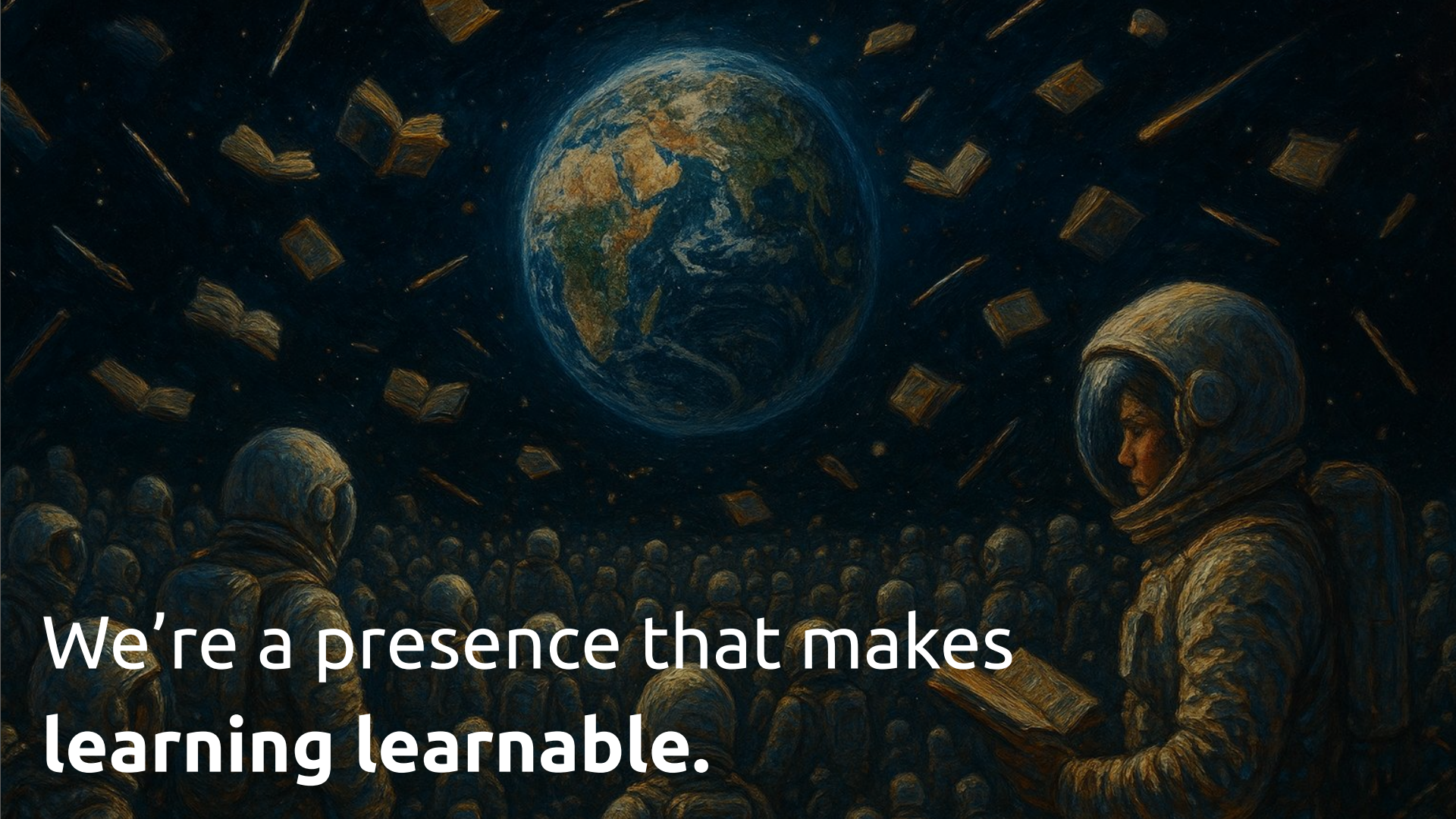
“EdTech”



They digitized **content**; not **consumption**.

Learners zone out, feel intimidated, and lose motivation—not from lack of ability, but because the content has **always** been passive, dense, unengaging, and, worst of all, fragmented.

And somehow, this is called “learning” –
but we’re flipping the script.



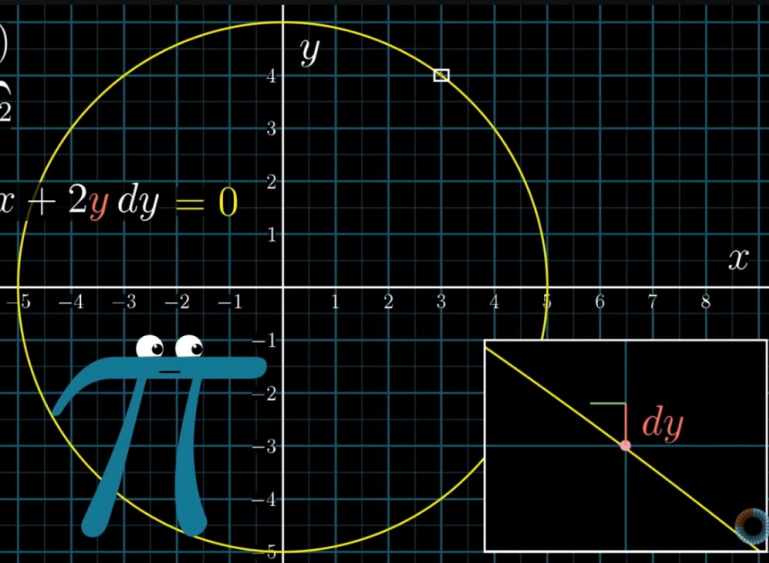
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learning learnable.

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product demo ...

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$$S(x, y) = x^2 + y^2$$
$$dS = 2x dx + 2y dy = 0$$



Implicit differentiation, what's going on here? | Chapter 6, Essence of calculus

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Reese's
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lim $h \rightarrow 0$
18:27
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Calculus 1 Lecture 2.7: Implicit Differentiation
Professor Leonard
553K views · 11 years ago

The Essence of Calculus
17:05
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42 - The implicit function theorem
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143K views · 9 years ago

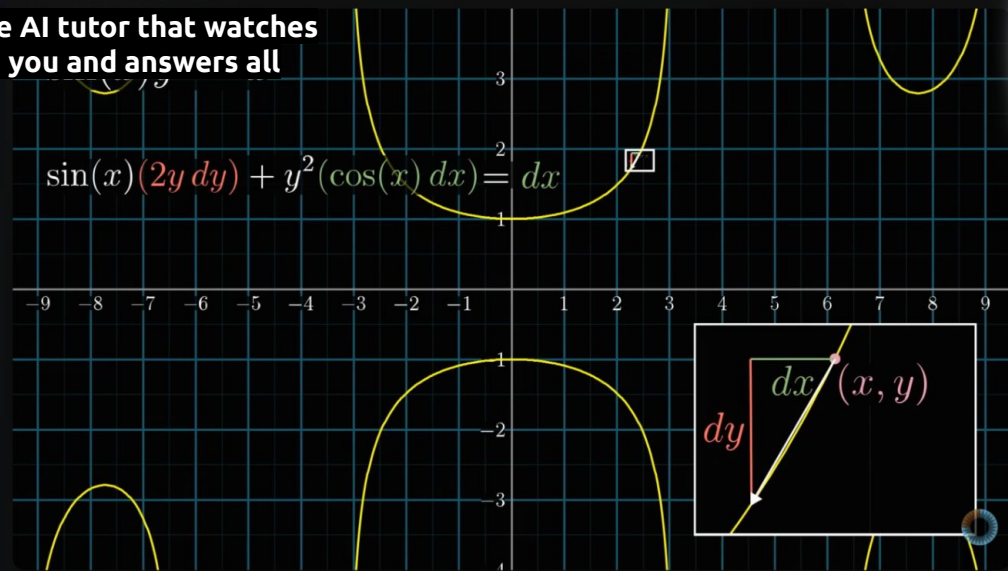
But what is a Laplace Transform?
3Blue1Brown  699K views · 10 days ago

Before Brekfuz



(let the content stick - ask it to help you practice and retain information!)

(context-aware AI tutor that watches the video with you and answers all your doubts)



Implicit differentiation, what's going on here? | Chapter 6. Essence of calculus

0:00

Open Brekfuz to Revise with Flashcards

Open Brekfuz to Review Notes

Open Brekfuz to Generate a Worksheet

(relevant questions with AI-graded personalized feedback at high-leverage moments.)

With Brekfuz

Key Concepts & Definitions

Related Rates

Related rates involve finding how fast one quantity changes using how fast another related quantity changes.

Example: As a balloon's radius grows, we find how fast its volume increases.

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Limits, L'Hôpital's rule, and epsilon-delta definitions

Implicitly Differentiate:

$\lim_{h \rightarrow 0} \frac{\sin(x)y^3 - x^2}{h}$

Reveal Hints (0/4 Revealed)

Reveal Next Hint

Differentiate LHS and RHS in isolation:

$\cos(x)y^3 + 3\sin(x)y^2 dy/dx = 2x$

Solve for dy/dx :

$dy/dx = (2x - \cos(x)y^3)/3\sin(x)y^2$

Check Answer

Practice Quiz

But what is a Laplace

15:33

(learn faster - ask it to intelligently play the video, removing certain topics, irrelevant information, and redundancies)

Ask me Anything!

Help me learn faster

Keeping important content

Removing unnecessary information

Maintaining Context

$S(x, y)$

$dS = 2x dx + 2y dy = 0$

y

x

π

dy

Introduction

Repeat Information

er 6, Essence

Self-Correction

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But what is a Laplace Transform?

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Calculus 1 Lecture 2.7: Implicit Differentiation

Professor Leonard

553K views · 11 years ago

1:08:11

The Essence of Calculus

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42 - The Implicit function theorem

Channel Updates

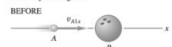
With Brekfuz

(thrive only on essential content and maximize your learning efficiency!)

250 CHAPTER 8. Momentum, Impulse, and Collisions

Figure 8.23 One-dimensional elastic collisions between objects with different masses.

(a) Moving Ping-Pong ball strikes initially stationary bowling ball.



(b) Moving bowling ball strikes initially stationary Ping-Pong ball.



(c) Moving Ping-Pong ball strikes initially stationary Ping-Pong ball.



(d) Moving Ping-Pong ball strikes initially stationary Ping-Pong ball.

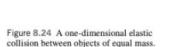
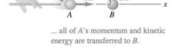


Figure 8.24 A one-dimensional elastic collision between objects of equal mass.

When a moving object A has a 1-D elastic collision with an equal mass, motionless object B ...



...all of A's momentum and kinetic energy are transferred to B.



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First we rearrange Eqs. (8.19) and (8.20) as follows:

$$m_B v_{B2i}^2 = m_A (v_{A1i}^2 - v_{A2i}^2) = m_A (v_{A1i} - v_{A2i})(v_{A1i} + v_{A2i}) \quad (8.21)$$

$$m_B v_{B2i} = m_A (v_{A1i} - v_{A2i}) \quad (8.22)$$

Now we divide Eq. (8.21) by Eq. (8.22) to obtain

$$v_{B2i} = v_{A1i} + v_{A2i} \quad (8.23)$$

We substitute this expression back into Eq. (8.22) to eliminate v_{B2i} , and then solve for v_{A2i} :

$$m_B (v_{A1i} + v_{A2i}) = m_A (v_{A1i} - v_{A2i}) \quad (8.24)$$

Finally, we substitute this result back into Eq. (8.23) to obtain

$$v_{B2i} = \frac{2m_A}{m_A + m_B} v_{A1i} \quad (8.25)$$

Now we can interpret the results. Suppose A is a Ping-Pong ball and B is a bowling ball. Then we expect A to bounce off after the collision with a velocity nearly equal to its original value but in the opposite direction (Fig. 8.23a), and we expect B's velocity to be much less. That's just what the equations predict. When m_A is much smaller than m_B , the fraction in Eq. (8.24) is approximately equal to -1 , so v_{A2i} is approximately equal to $-v_{A1i}$. The fraction in Eq. (8.25) is much smaller than unity, so v_{B2i} is much less than v_{A1i} . Figure 8.23b shows the opposite case, in which A is the bowling ball and B the Ping-Pong ball and m_A is much larger than m_B . What do you expect to happen then? Check your predictions against Eqs. (8.24) and (8.25).Another interesting case occurs when the masses are equal (Fig. 8.24). If $m_A = m_B$, then Eqs. (8.24) and (8.25) give $v_{A2i} = 0$ and $v_{B2i} = v_{A1i}$. That is, the object that was moving stops it; it gives all its momentum and kinetic energy to the object that was at rest. This behavior is familiar to all pool players.

Elastic Collisions and Relative Velocity

Let's return to the more general case in which A and B have different masses. Equation (8.23) can be rewritten as

$$v_{B1i} - v_{A1i} = v_{B2i} - v_{A2i} \quad (8.26)$$

Here $v_{B1i} - v_{A1i}$ is the velocity of B relative to A after the collision; from Eq. (8.26), this equals v_{A1i} , which is the negative of the velocity of A relative to B before the collision. (We discussed relative velocity in Section 3.5.) The relative velocity has the same magnitude, but opposite sign, before and after the collision. The sign changes because A and B are approaching each other before the collision but moving apart after the collision. If we view this collision from a second coordinate system moving with constant velocity relative to the first, the velocities of the objects are different but the relative velocities are the same. Hence our statement about relative velocities holds for any straight-line elastic collision, even when neither object is at rest initially. *In a straight-line elastic collision of two objects, the relative velocities before and after the collision have the same magnitude but opposite sign.* This means that if B is moving before the collision, Eq. (8.26) becomes

$$v_{B2i} - v_{A2i} = -(v_{B1i} - v_{A1i}) \quad (8.27)$$

It turns out that a vector relationship similar to Eq. (8.27) is a general property of all elastic collisions, even when both objects are moving initially and the velocities do not all lie along the same line. This result provides an alternative and equivalent definition of an elastic collision: *In an elastic collision, the relative velocity of the two objects has the same magnitude before and after the collision.* Whenever this condition is satisfied, the total kinetic energy is also conserved.

When an elastic two-object collision isn't head-on, the velocities don't all lie along a single line. If they all lie in a plane, then each final velocity has two unknown components, and there are four unknowns in all. Conservation of energy and conservation of the x- and y-components of momentum give only three equations. To determine the final velocities uniquely, we need additional information, such as the direction or magnitude of one of the final velocities.

EXAMPLE 8.10 An elastic straight-line collision

We repeat the air-track collision of Example 8.5 (Section 8.2), but now we add ideal spring bumpers to the gliders so that the collision is elastic. What are the final velocities of the gliders?

IDENTIFY and SET UP The net external force on the system is zero, so the momentum of the system is conserved. Figure 8.25 shows our sketch. We'll find our target variables, v_{A2i} and v_{B2i} , by using Eq. (8.27), the relative-velocity relationship for an elastic collision, and the momentum-conservation equation.

EXECUTE From Eq. (8.27),

$$v_{B2i} - v_{A2i} = -(v_{B1i} - v_{A1i}) \\ = -(-2.0 \text{ m/s} - 2.0 \text{ m/s}) = 4.0 \text{ m/s}$$

From conservation of momentum,

$$m_A v_{A1i} + m_B v_{B1i} = m_A v_{A2i} + m_B v_{B2i} \\ (0.50 \text{ kg})(2.0 \text{ m/s}) + (0.30 \text{ kg})(-2.0 \text{ m/s}) \\ = (0.50 \text{ kg})v_{A2i} + (0.30 \text{ kg})v_{B2i} \\ 0.50 v_{A2i} + 0.30 v_{B2i} = 0.40 \text{ m/s}$$

(To get the last equation we divided both sides of the equation just above it by 1 kg. This makes the units the same as in the first equation.) Solving these equations simultaneously, we find

$$v_{A2i} = -1.0 \text{ m/s} \quad v_{B2i} = 3.0 \text{ m/s}$$

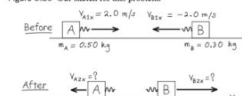
EVALUATE Both objects reverse their direction of motion. A moves to the left at 1.0 m/s and B moves to the right at 3.0 m/s. This is unlike the result of Example 8.5 because that collision was *not* elastic. The more massive glider A slows down in the collision and so loses kinetic energy.

EXAMPLE 8.11 Moderating fission neutrons in a nuclear reactor

The fission of uranium nuclei in a nuclear reactor produces high-speed neutrons. Before such neutrons can efficiently cause additional fissions, they must be slowed down by collisions with nuclei in the *moderator* of the reactor. The first nuclear reactor (built in 1942 at the University of Chicago) used carbon (graphite) as the moderator. Suppose a neutron (mass 1 u) traveling at $2.6 \times 10^3 \text{ m/s}$ undergoes a head-on elastic collision with a carbon nucleus (mass 12 u) initially at rest. Neglecting external forces during the collision, find the velocities after the collision. (1 u is the atomic mass unit, equal to $1.66 \times 10^{-27} \text{ kg}$.)

IDENTIFY and SET UP We ignore external forces, so momentum is conserved in the collision. The collision is elastic, so kinetic energy is also conserved. Figure 8.26 shows our sketch. We take the x-axis to be in the direction in which the neutron is moving initially. The collision is head-on, so both particles move along this same axis after the collision. The carbon nucleus is initially at rest, so we can use Eqs. (8.24) and (8.25); we replace A by n (for the neutron) and B by C (for the carbon nucleus). We have $m_n = 1.0 \text{ u}$, $m_C = 12 \text{ u}$, and $v_{C1i} = 2.6 \times 10^3 \text{ m/s}$. The target variables are the final velocities v_{n2i} and v_{C2i} .

Figure 8.25 Our sketch for this problem.



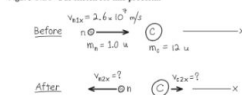
The less massive glider B speeds up and gains kinetic energy. The total kinetic energy before the collision (which we calculated in Example 8.7) is 1.6 J. The total kinetic energy after the collision is

$$\frac{1}{2}(0.50 \text{ kg})(-1.0 \text{ m/s})^2 + \frac{1}{2}(0.30 \text{ kg})(3.0 \text{ m/s})^2 = 1.6 \text{ J}$$

The kinetic energies before and after this elastic collision are equal. Kinetic energy is transferred from A to B, but none of it is lost.

CAUTION Be careful with the elastic collision equations. You could not have solved this problem by using Eqs. (8.24) and (8.25), which apply only if object B is initially at rest. Always be sure that you use equations that are applicable!

Figure 8.26 Our sketch for this problem.



EXECUTE You can do the arithmetic. (Hint: There's no reason to convert atomic mass units to kilograms.) The results are

$$v_{n2i} = -2.2 \times 10^3 \text{ m/s} \quad v_{C2i} = 0.4 \times 10^3 \text{ m/s}$$

Continued

Figure 8.23 One-dimensional elastic collision between objects with different masses.

(a) Ping-Pong ball strikes initially stationary bowling ball.

BEFORE

AFTER

(b) Moving bowling ball strikes initially stationary Ping-Pong ball.

BEFORE

AFTER

Figure 8.24 A one-dimensional elastic collision between objects of equal mass.

When a moving object A has a 1-D elastic collision with an equal-mass, motionless object B ...

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We substitute this expression back into Eq. (8.22) to eliminate v_{B2} , and then solve for v_{A2} :

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$$v_{A2} = \frac{m_A - m_B}{m_A + m_B} v_{A1}$$

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EXECUTE: From Eq. (8.27),

$$v_{B2} - v_{A2} = -(v_{B1} - v_{A1}) = -(-2.0 \text{ m/s} - 2.0 \text{ m/s}) = 4.0 \text{ m/s}$$

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$$\begin{aligned} m_A v_{A1} + m_B v_{B1} &= m_A v_{A2} + m_B v_{B2} \\ (0.50 \text{ kg})(2.0 \text{ m/s}) + (0.30 \text{ kg})(-2.0 \text{ m/s}) &= (0.50 \text{ kg})v_{A2} + (0.30 \text{ kg})v_{B2} \\ 0.50 v_{A2} + 0.30 v_{B2} &= 0.40 \text{ m/s} \end{aligned}$$

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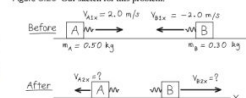
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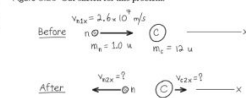
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Figure 8.26 Our sketch for this problem.



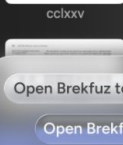
Key Concepts & Definitions

Elastic Collisions

In an elastic collision both total momentum and total kinetic energy are conserved. The relative velocity of the two colliding objects has the same magnitude after the collision as before, but in the opposite direction.

(key concepts and definitions
stack - no need to keep
revisiting the entire book!)

(context-aware AI tutor that helps you exactly in the way you've studied; no new out-of-syllabus BS)



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Open Brekfuz to Revise with Flashcards

15th Edition By Hugh D. Young_c...



(practice question with hints & i ask the tutor to quiz you!)

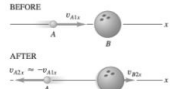
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With Brekfuz

250 CHAPTER 8 Momentum, Impulse, and Collisions

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(b) Moving bowling ball strikes initially stationary Ping-Pong ball.

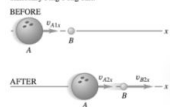
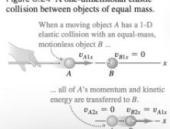


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$$v_{B2i} = v_{A1i} + v_{A2i} \quad (8.23)$$

We substitute this expression back into Eq. (8.22) to eliminate v_{B2i} , and then solve for v_{A2i} :

$$m_B (v_{A1i} + v_{A2i}) = m_A (v_{A1i} - v_{A2i}) \quad (8.24)$$

Finally, we substitute this result back into Eq. (8.23) to obtain

$$v_{B2i} = \frac{2m_A}{m_A + m_B} v_{A1i} \quad (8.25)$$

Now we can interpret the results. Suppose A is a Ping-Pong ball and B is a bowling ball. Then we expect A to bounce off after the collision with a velocity nearly equal to its original value but in the opposite direction (Fig. 8.23a), and we expect B 's velocity to be much less. That's just what the equations predict. When m_A is much smaller than m_B , the fraction in Eq. (8.24) is approximately equal to -1 , so v_{A2i} is approximately equal to $-v_{A1i}$. The fraction in Eq. (8.25) is much smaller than unity, so v_{B2i} is much less than v_{A1i} . Figure 8.23b shows the opposite case, in which A is the bowling ball and B the Ping-Pong ball and m_A is much larger than m_B . What do you expect to happen then? Check your predictions against Eqs. (8.24) and (8.25).

Another interesting case occurs when the masses are equal (Fig. 8.24). If $m_A = m_B$, then Eqs. (8.24) and (8.25) give $v_{A2i} = 0$ and $v_{B2i} = v_{A1i}$. That is, the object that was moving stops dead; it gives all its momentum and kinetic energy to the object that was at rest. This behavior is familiar to all pool players.

Elastic Collisions and Relative Velocity

Let's return to the more general case in which A and B have different masses. Equation (8.23) can be rewritten as

$$v_{A1i} = v_{B2i} - v_{A2i} \quad (8.26)$$

Here $v_{B2i} - v_{A2i}$ is the velocity of B relative to A after the collision; from Eq. (8.26), this equals v_{A1i} , which is the negative of the velocity of B relative to A before the collision. (We discussed relative velocity in Section 3.5.) The relative velocity has the same magnitude, but opposite sign, before and after the collision. The sign changes because A and B are approaching each other before the collision but moving apart after the collision. If we view this collision from a second coordinate system moving with constant velocity relative to the first, the velocities of the objects are different but the relative velocities are the same. Hence our statement about relative velocities holds for any straight-line elastic collision, even when neither object is at rest initially. **In a straight-line elastic collision of two objects, the relative velocities before and after the collision have the same magnitude but opposite sign.** This means that if B is moving before the collision, Eq. (8.26) becomes

$$v_{B2i} - v_{A2i} = -(v_{B1i} - v_{A1i}) \quad (8.27)$$

It turns out that a vector relationship similar to Eq. (8.27) is a general property of all elastic collisions, even when both objects are moving initially and the velocities do not all lie along the same line. This result provides an alternative and equivalent definition of an elastic collision: **In an elastic collision, the relative velocity of the two objects has the same magnitude before and after the collision.** Whenever this condition is satisfied, the total kinetic energy is also conserved.

When an elastic two-object collision is line. If they all lie in a plane, then each is are four unknowns in all. Conservation of of momentum give only three equations. additional information, such as the direction of the velocity of the objects after the collision.

EXAMPLE 8.10 An elastic straight-line

We repeat the air-track collision of Example 8.9 with different masses. Suppose the gliders have masses $m_A = 2.0 \text{ kg}$ and $m_B = 1.0 \text{ kg}$. What are the final velocities of the gliders?

IDENTIFY and SET UP: The net external force on the system is zero, so the momentum of the system is conserved. We'll find our target variables, v_{A2i} and v_{B2i} , using the relative-velocity relationship for an elastic collision.

EXECUTE: From Eq. (8.27),

$$v_{B2i} - v_{A2i} = -(v_{B1i} - v_{A1i})$$

$$v_{B2i} - v_{A2i} = -(-2.0 \text{ m/s} - 2.0 \text{ m/s})$$

$$v_{B2i} - v_{A2i} = 4.0 \text{ m/s}$$

$$v_{B2i} = v_{A2i} + 4.0 \text{ m/s}$$

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$$v_{B2i} = v_{A2i} + 4.0 \text{ m/s}$$

Practice Question

Cart A weighs **2 kgs** and is moving at a velocity of **+3 m/s**. Cart B weighs **1 kg** and is moving at a velocity of **-2 m/s**. After coming in contact, they stick together.

Find the **final velocity** of the carts.

$$\begin{aligned} \text{Initial momentum} &= 2 \times 3 + 1 \times (-2) = 4 \text{ kg m/s} \\ \text{Final Mass} &= 1 + 2 = 3 \text{ kg} \end{aligned}$$

$$\begin{aligned} \text{Therefore, since Final Momentum} &= \text{Initial Momentum:} \\ 4 &= 3v \\ v &= 1.33 \text{ m/s} \end{aligned}$$



Correctly Calculated Initial Momentum

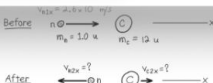


Correctly Identified the Final Mass is the Sum of Carts



Correctly computed final velocity with Momentum

Check Answer



Key Concepts & Definitions

Elastic Collisions

In an elastic collision both total momentum and total kinetic energy are conserved. The relative velocity of the two colliding objects has the same magnitude after the collision as before, but in the opposite direction.

An artistic illustration of three astronauts in space suits floating in a dark blue space filled with stars. A large, bright orange sphere, resembling a sun or a planet, is in the upper left. The astronaut on the left is holding a small object. The middle astronaut is holding a small orange sphere. The astronaut on the right is holding a small object. The text "Brekfuz • What if studying finally worked?" is overlaid in the center.

Brekfuz • What if studying
finally worked?